Retail Sales Performance Analysis

Insights Report

Introduction

- This project aims to analyze retail sales performance using Business Intelligence techniques.
  The analysis is based on transactional sales data from a Superstore dataset, including information about customers, products, regions, dates, sales, and profit.
- A star schema data model was designed, and an interactive Power BI dashboard was developed to support decision-making through an Executive Summary and a Deep Dive analysis.
- Tool used: **Power BI**
- Objective: **support managerial decisions**

Key Findings

Insight 1 — Profitability by Product Category

- ***Key Finding***

The **Technology** category is the most profitable product category, generating the highest total profit, followed by **Office Supplies**, while **Furniture** contributes significantly less profit.

- ***Evidence from Dashboard***

This insight is derived from the *"Profit by Product Category"* bar chart, where Technology shows the longest bar, Office Supplies ranks second, and Furniture displays a much smaller profit contribution.

- ***Why This Matters***

High profitability in the Technology category indicates stronger margins or more efficient cost structures. Conversely, the relatively low profitability of Furniture suggests potential issues such as high production, logistics, or discounting costs. This insight helps management prioritize profitable categories rather than focusing solely on sales volume.

Key Findings

Insight 2 — Profit Distribution Across Sub-Categories

- ***Key Finding***

Profitability varies significantly across sub-categories. **Copiers**, **Phones**, and **Accessories** generate the highest profits, while **Tables** show a **negative profit**, indicating losses.

- ***Evidence from Dashboard***
- The *"Profit by Sub-Category"* chart shows Copiers as the top-performing sub-category, while
- Tables appear below zero on the profit axis, confirming negative profitability.

- ***Why This Matters***

Not all products contribute positively to company performance. Loss-making sub-categories reduce overall profitability and may require corrective actions such as price adjustments, cost reduction, or discontinuation.

Key Findings

Insight 3 — Sales Trend Over Time

- ***Key Finding***

Total sales exhibit a **strong upward trend** from 2015 to 2017, following a slight decline between 2014 and 2015.

- ***Evidence from Dashboard***

The *"Sales Trend Over Time"* line chart shows a decrease in sales in 2015, followed by a significant and continuous increase in 2016 and 2017.

- ***Why This Matters***

The positive sales trajectory indicates business growth and increasing market demand. Identifying the turning point after 2015 allows management to analyze successful strategies implemented during this period and replicate them to sustain long-term growth.

Business Recommendations

- **Prioritize Investment in the Technology Category**

The company should increase investment in the Technology category by expanding product offerings, strengthening supplier relationships, and allocating more marketing resources.

- **Improve Profitability in the Furniture Category**

Management should conduct a cost and pricing review of Furniture products, including logistics costs, supplier pricing, and discount strategies.

- **Leverage Growth Momentum for Expansion**

The company should use the positive sales trend to support expansion strategies such as entering new regions, launching new products, or increasing production capacity.

- **Use Historical Trends for Demand Forecasting**

Management should integrate sales trend analysis into forecasting and inventory planning processes.

Limitations & Future Improvements

Limitations of the Analysis

- **1. Historical Data Limitation**

The analysis is based on historical sales data, which may not accurately reflect current market conditions or future trends.

- **2. Lack of Detailed Cost Information**

Profit is analyzed at an aggregated level, without detailed cost components, limiting precise identification of profitability drivers.

- **3. Limited Customer Information**

The absence of detailed customer demographic and behavioral data restricts deeper customer segmentation analysis.

- **4. External Factors Not Considered**

External factors such as economic conditions and competition are not included, although they may influence sales and profit performance.

- **5. Product-Level Granularity Constraints**

Some detailed operational information is missing, limiting more detailed product-level analysis.

FUTURE IMPROVEMENTS

To enhance the quality and strategic value of the BI solution, the following improvements are recommended:

- **Integration of Real-Time or Updated Data Sources**
  Connecting Power BI to live databases would allow continuous monitoring of sales and profitability, enabling faster and more informed decision-making.
- **Inclusion of Detailed Cost and Operational Data**
  Adding cost breakdowns (logistics, procurement, storage) would support more accurate profitability analysis and help explain losses in certain sub-categories.
- **Advanced Customer Analytics**
  Incorporating customer demographics and behavioral metrics would allow segmentation, churn analysis, and targeted marketing strategies.
- **Predictive Analytics and Forecasting**
  Implementing sales forecasting models would help anticipate demand trends and support inventory and capacity planning.
- **Enhanced Interactivity and Drill-Through Analysis**
  Adding drill-through pages and more interactive visuals would allow users to explore performance at product, region, or customer levels in greater detail.
- **Benchmarking and External Data Integration**
  Integrating external market or competitor data would provide context and help evaluate performance relative to industry benchmarks.